

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2nd Semester of B. Pharm. Examination

University Theory Examination April-May 2015

PH109 Pharmaceutical Engineering - II

Date: 13/05/2015, Wednesday Time: 10:00 a.m to 1:00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

- A Define: Filtration, Write a short note on mechanism of filtration and filter-aids. 10
- B Explain principle, working, advantages and disadvantages with labeled diagram of Plate & Frame filter press. How plate and frame filter press can be modified? 10
- C a) Define: Centrifugation, Discuss principle and theory of centrifugation. 05
b) Write a short note on: Perforated basket centrifuge 05

Q 2 Attempt any TWO of the following

- A a) Write a short note: drying rate curve. 05
b) Write a short note: Fluidized bed dryer 05
- B Define: Lyophilization, Explain in brief about the freeze dryer with schematic diagram, principle, construction, working, advantages, disadvantages and uses. 10
- C a) Write a short note on Spray dryer 05
b) Write a short note on factors affecting filtration process. 05

SECTION- II

Q 3 Attempt any TWO of the following

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| A | What are the factors affect the rate of evaporation? | 10 |
| B | a) Write a note on falling film evaporator. | 05 |
| | b) Explain the humidity chart | 05 |
| C | Whrite a note on azeotropic distillation | 10 |

Q 4 Attempt any TWO of the following

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| A | What is the application of HVAC system in pharmaceutical industry ? | 10 |
| B | a) Classify the condenser | 05 |
| | b) Giver the note on rectification | 05 |
| C | Write a note on evaporating stills | 10 |
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